

SAMPLING COHERENT WEATHER FIELDS

FROM PROBABILISTIC FOUNDATION MODELS

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Word count: 990 Limit: 1,000 words

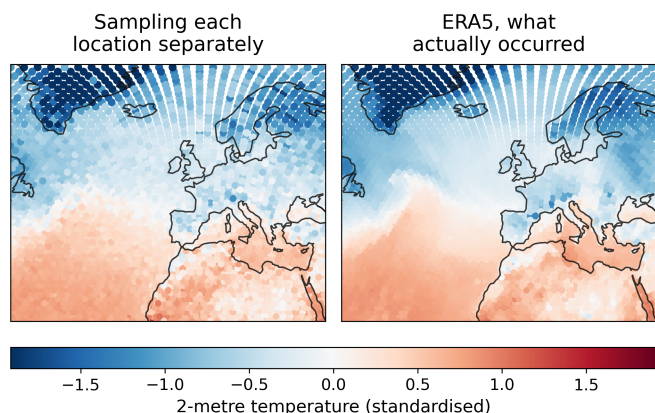


Figure 1: A field drawn independently from each location's emitted mixture (left) carries fine-scale noise that no real weather field shows, whereas the ERA5 [5] record of 2-metre temperature on the same grid (right) varies coherently across space. Each dot is one grid location, shown over the North Atlantic and Europe.

Motivation & Scientific Justification

Weather forecasts inform decisions across agriculture, energy, transport and emergency planning, and with recent years repeatedly breaking temperature records, this means preparing for droughts and heatwaves depends not only on the most likely outcome but on how confident that prediction is. Operational ensembles traditionally estimate this uncertainty through repeated physics-based simulations that vary initial conditions and model behaviour. Machine-learning models offer a newer route, learning from decades of past atmospheric data without directly simulating the physics. Some, such as GenCast [1], generate whole ensembles of fields. The WeatherGenerator [2], a foundation weather model developed at ECMWF is related to AtmoRep [3] and Aurora [4]. Its branch instead outputs a Gaussian mixture at every grid location, a blend of four weighted bell-shaped components. Each mixture can therefore represent uncertainty and hold several distinct plausible values at once. However, due to computational constraints each location's mixture is produced on its own, and the model holds no covariance between locations. Sampling each location independently therefore gives a noisy field unlike any real weather pattern (Figure 1). Many of the quantities forecasts exist to provide belong to a whole field rather than a single location, such as the total rainfall over a river catchment, and each depends on how locations vary together, so it can only be drawn from a field that could actually occur. This research supplies that missing step for this Gaussian-mixture branch, assembling its independently produced mixtures into one coherent field.

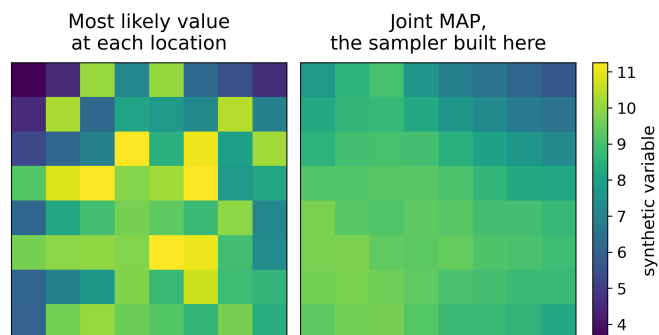


Figure 2: On the synthetic testbed, taking each location's most likely value leaves a patchy field (left) because neighbouring locations rarely agree on one value, whereas the sampler recovers a smooth field (right) while keeping each value plausible.

Methodology

The sampling method, Joint MAP, chooses one value at every location by balancing two requirements. Each chosen value must stay plausible under its own location's mixture, and neighbouring values must vary smoothly across the grid. The smoothness is a prior we assume and impose on the method, not a recovery of the covariance the model discarded. The research takes the emitted mixtures as fixed input. It is not itself a forecasting model and does not aim to improve forecast accuracy. The strength of the smoothing can be varied to suit the use. For comparison it is tuned to a fixed reference, so competing methods are judged at the same smoothness, with no setting treated as correct. The sampler was first exercised on a small synthetically constructed testbed whose mixtures are specified in advance, so the coherent underlying distributions are known and give context on how successfully the sampler rebuilds a field. Its independently drawn mixture weights also make it a demanding case, with few locations holding one dominant value, so the field of most likely values is patchy and only a joint choice recovers a smooth one (Figure 2). The sampler was then tested on real 48-hour temperature forecasts across roughly forty thousand grid locations covering the globe, from 12 start dates spanning the seasons of 2023, outside the model's training period.

Key Findings

The sampled fields were compared against several baselines, from an independent draw at each location to a smoothed field of most likely values. At the same smoothness the sampler was found to keep far more locations on values the model itself rates plausible, with the smoothed baseline leaving 32.7% of locations on unlikely in-between values against the sampler's 13.2%, roughly two and a half times as many. The advantage concentrates where the forecast is genuinely uncertain, at locations whose mixture holds two distinct outcomes separated by a low-probability gap, a front arriving or not, rather than one wide overlapping spread. Within those locations the smoothed base-

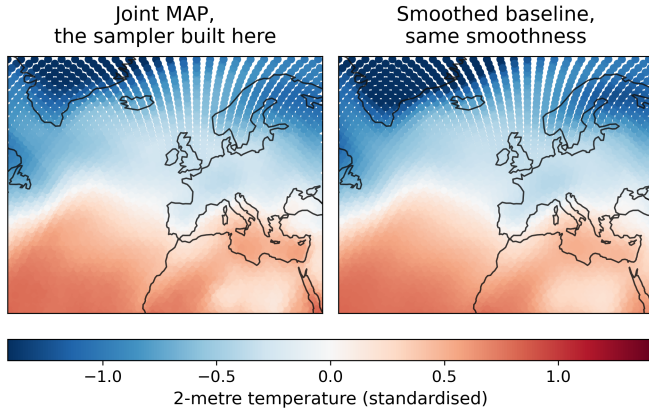


Figure 3: Sampled 48-hour temperature fields for one 2023 start date over the North Atlantic and Europe, in standardised units, matched in smoothness by construction. To the eye the two are alike, so the comparison is decided by the share of values each leaves on outcomes the model itself rated unlikely.

line pushes 19.3% of values into the unlikely middle ground between the two outcomes (95% interval [17.8%, 20.8%]) against the sampler’s 9.0% ([7.9%, 10.1%]), an ordering that held on all 12 start dates. Additionally, nearly every value the sampler selects sits centrally within its own mixture, 99.1% of locations falling in the central 90% band ([98.9%, 99.3%] across start dates). This checks each value against its own predicted distribution, not the calibration of forecast probabilities against observed weather. Such situations become more common further ahead. Locations effectively certain of one value fall from 67.9% at six hours to 9.3% at 48 hours, whilst locations holding two well-separated outcomes grow from 2.5% to 6.3%, so the conditions where the sampler matters most grow with range. Figure 3 shows the sampler’s field beside the smoothed baseline at the same smoothness.

Impact & Limitations

This research provides a sampling method and evaluation architecture for WeatherGenerator’s Gaussian-mixture branch that were not previously built, both applying to any future model on the branch. More broadly, a coherent field is what lets this style of forecast answer the field-level questions raised above. The full evaluation runs in about 77 seconds on one processor. This research uses only one weather variable, 2-metre temperature, one spatial resolution and one model branch. The model behind the mixtures is an early checkpoint that was still visibly improving when training stopped, forecasts reach only 48 hours ahead, and the robustness check spans the seasons of a single test year. Every comparison is made against the model’s own emitted mixtures, the only ground truth available, and whether the sampled fields match the observed atmosphere is deliberately left to future work.

Next Steps

The first step is to train the model out to seven to fourteen days, where the mixtures should soften further. A preliminary seven-day run showed steady softening but no growth yet in well-separated alternatives, though artefacts of longer forecasts are difficult to separate from undertraining. The sampler’s

margin over the baselines should grow at these ranges if the mixtures keep genuinely distinct outcomes. Sampling several weather variables jointly, such as rain with cloud cover, would also make established methods like Ensemble Copula Coupling [6] and the Schaake shuffle [7] comparable. Finer-resolution mixtures, at points a few tens of kilometres apart, should vary more gently between neighbours, whilst weather fronts and other sharp features, which the sampler pushes against, need preserving. Validating the fields against observed weather is the further goal, a step towards forecasts that carry their uncertainty through to the decisions they inform.

References

- [1] Ilan Price, Alvaro Sanchez-Gonzalez, Ferran Alet, et al. Probabilistic weather forecasting with machine learning. *Nature*, 637(8044):84–90, 2025. doi: 10.1038/s41586-024-08252-9. Introduces the GenCast model; preprint arXiv:2312.15796.
- [2] WeatherGenerator Consortium. WeatherGenerator: Advancing earth system science with innovative AI. <https://weathergenerator.eu>; source code at <https://github.com/ecmwf/WeatherGenerator>, 2024. ECMWF-coordinated European initiative (EU Horizon grant 101187947). Accessed 2026-06-19.
- [3] Christian Lessig, Ilaria Luise, Bing Gong, Michael Langguth, Scarlet Stadler, and Martin Schultz. AtmoRep: A stochastic model of atmosphere dynamics using large scale representation learning. *arXiv preprint arXiv:2308.13280*, 2023.
- [4] Cristian Bodnar, Wessel P. Bruinsma, Ana Lucic, et al. A foundation model for the earth system. *Nature*, 641(8065): 1180–1187, 2025. doi: 10.1038/s41586-025-09005-y.
- [5] Hans Hersbach, Bill Bell, Paul Berrisford, et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. doi: 10.1002/qj.3803.
- [6] Roman Schefzik, Thordis L. Thorarinsdottir, and Tilman Gneiting. Uncertainty quantification in complex simulation models using ensemble copula coupling. *Statistical Science*, 28(4):616–640, 2013. doi: 10.1214/13-STS443.
- [7] Martyn Clark, Subhrendu Gangopadhyay, Lauren Hay, Balaji Rajagopalan, and Robert Wilby. The Schaake shuffle: A method for reconstructing space–time variability in forecasted precipitation and temperature fields. *Journal of Hydrometeorology*, 5(1):243–262, 2004.